Diffusion Models

DSAIT4030 Generative Modelling | Week 6 Lecture 1

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Recap: Score-based Models

- Score function: points in the direction of the increasing probability.
- We use a model $s_\theta(x) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ to learn the score function.

$$s_\theta(x) \approx \nabla_x \log\, p(x)$$

- Score matching: Training the model using data

$$\frac{1}{2}\mathbb{E}_{x\sim p}\left[\|s_\theta(x) - \nabla_x \log\, p(x)\|^2\right]$$

$$\equiv \frac{1}{2}\mathbb{E}_{x\sim p(x)}\left[s_\theta(x)^2\right] + \mathbb{E}_{x\sim p(x)}\left[\text{tr}\left(\nabla_x s_\theta(x)\right)\right]$$

- Issues: (1) Not Scalable. (2) Inaccurate in low density regions.

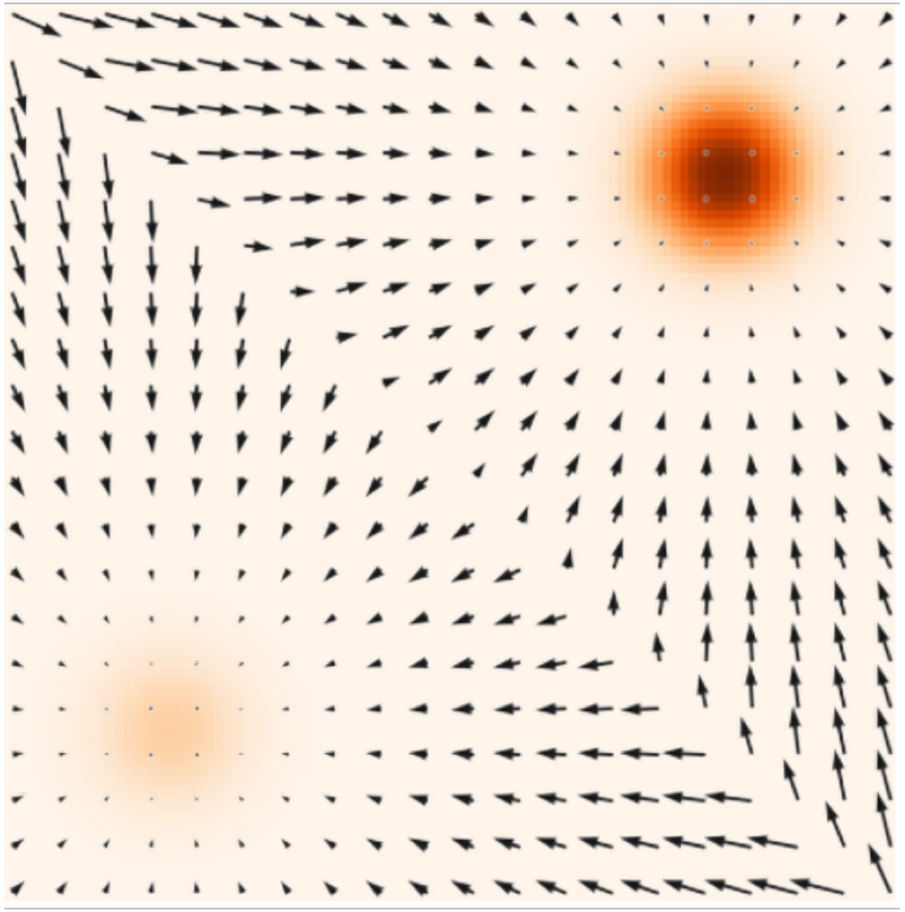


Image Source: Blog on Score-based Generative Models — Yang Song

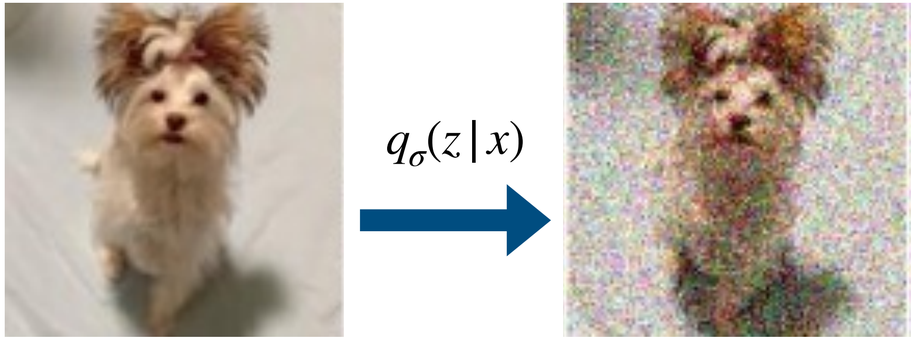
Recap: Denoising Score Matching

$$\frac{1}{2}\mathbb{E}_{x\sim p}\left[\|s_\theta(x) - \nabla_x \log\, p(x)\|^2\right]$$ **Original Objective**

$$\frac{1}{2}\mathbb{E}_{z\sim q_\sigma}\left[\|s_\theta(z) - \nabla_z \log\, q_\sigma(z)\|^2\right]$$ **Noise-perturbed Objective**

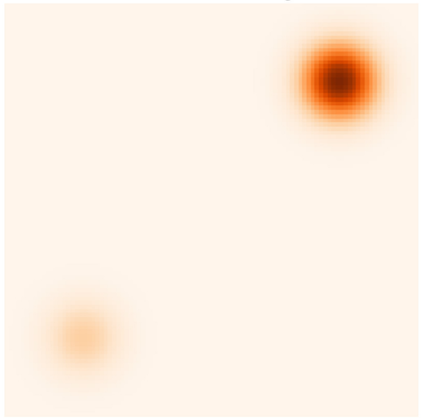
$$\equiv \frac{1}{2}\mathbb{E}_{x\sim p,\, z\sim q_\sigma(z|x)}\left[\|s_\theta(z) - \nabla_z \log\, q_\sigma(z\,|\,x)\|^2\right]$$

$$\equiv \frac{1}{2}\mathbb{E}_{x\sim p,\, z\sim q_\sigma(z|x)}\left[\|s_\theta(z) + \frac{\epsilon}{\sigma^2}\|^2\right]$$

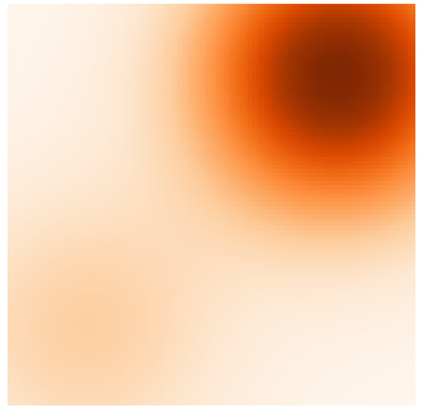


$q_\sigma(z\,|\,x)$

$x \sim p(x)$ $\qquad$ $z \sim q_\sigma(z)$



$p(x)$

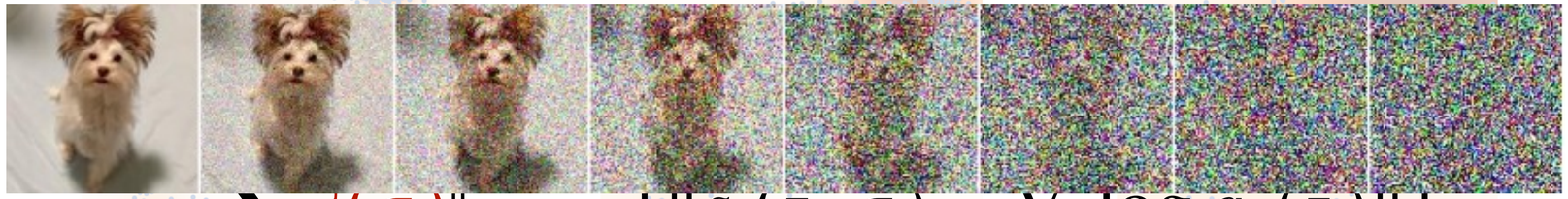


$$q_\sigma(z) = \int q(z\,|\,x,\sigma)p(x)dx$$

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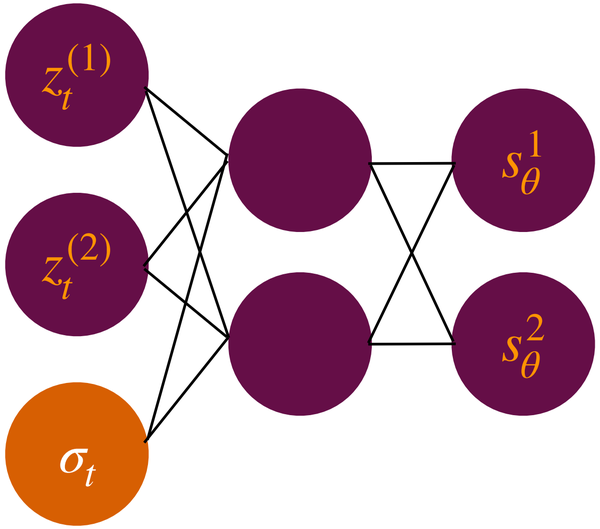
Recap: Joint Score Estimation

$$\sigma_1 < \sigma_2 < \sigma_3$$



$$\frac{1}{T}\sum_{t=1}^{T} \lambda(\sigma_t)\mathbb{E}_{z_t \sim q_{\sigma_t}}\left[\|s_\theta(z_t, \sigma_t) - \nabla_x \log q_{\sigma_t}(z_t)\|\right]$$

weighing function



Noise Conditional Score Network

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Image Source: Blog on Score-based Generative Models — Yang Song

Recap: Annealed Langevin Dynamics

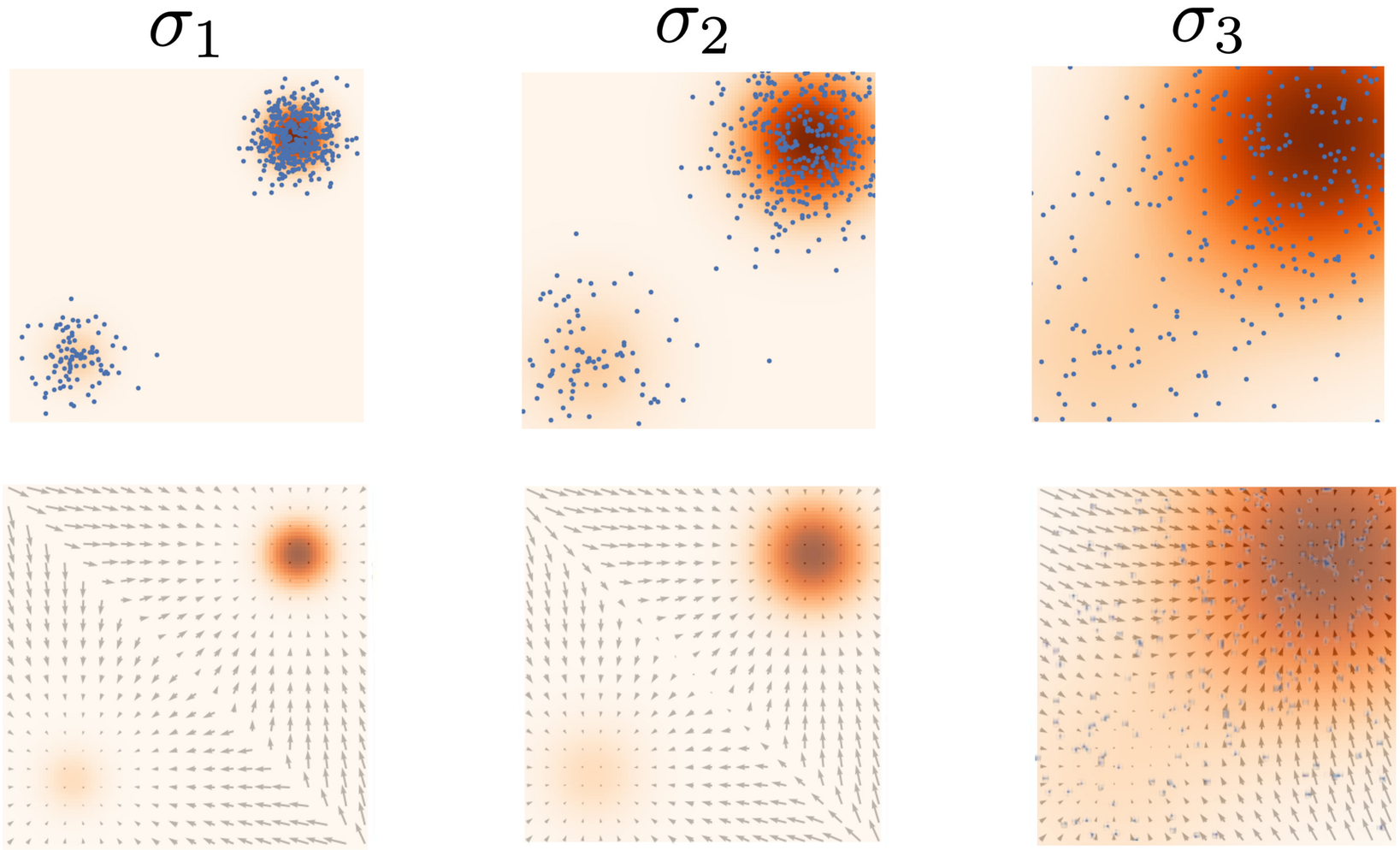
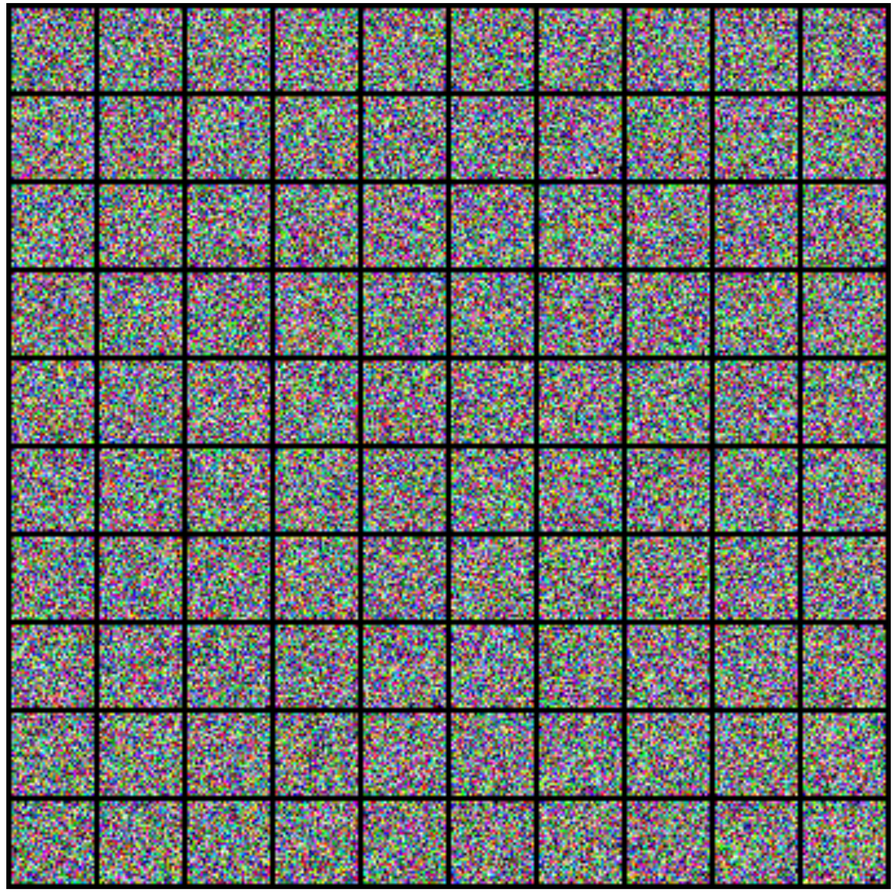
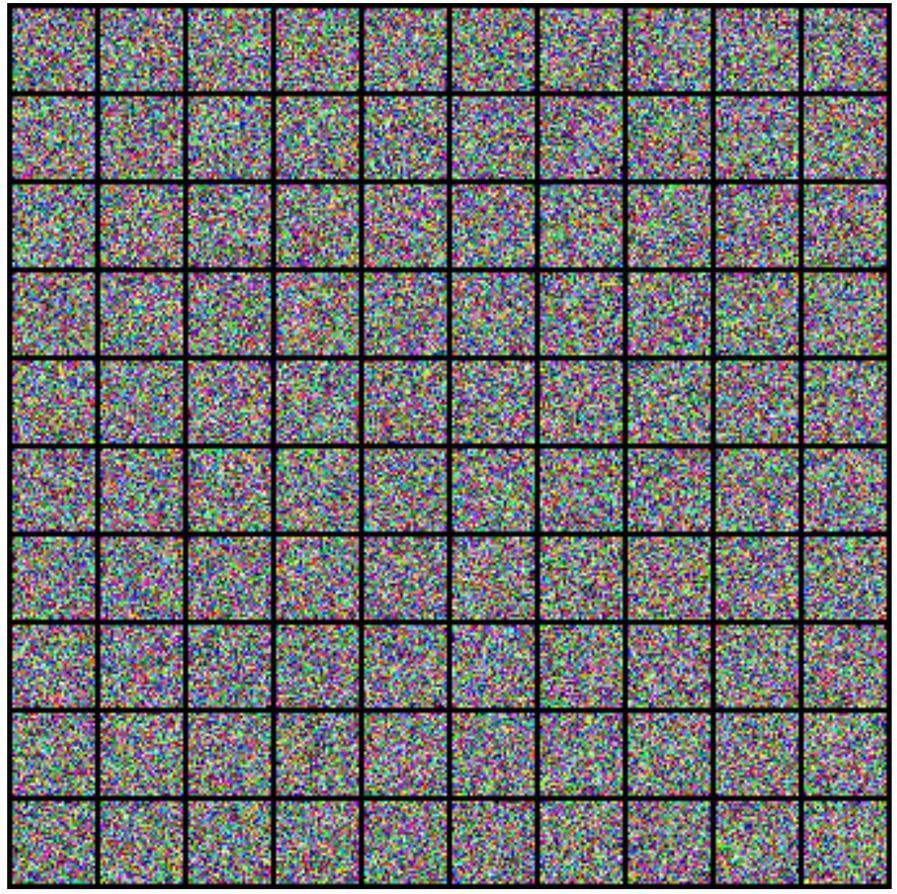


Image Source: Blog on Score-based Generative Models — Yang Song

Recap: Annealed Langevin Dynamics

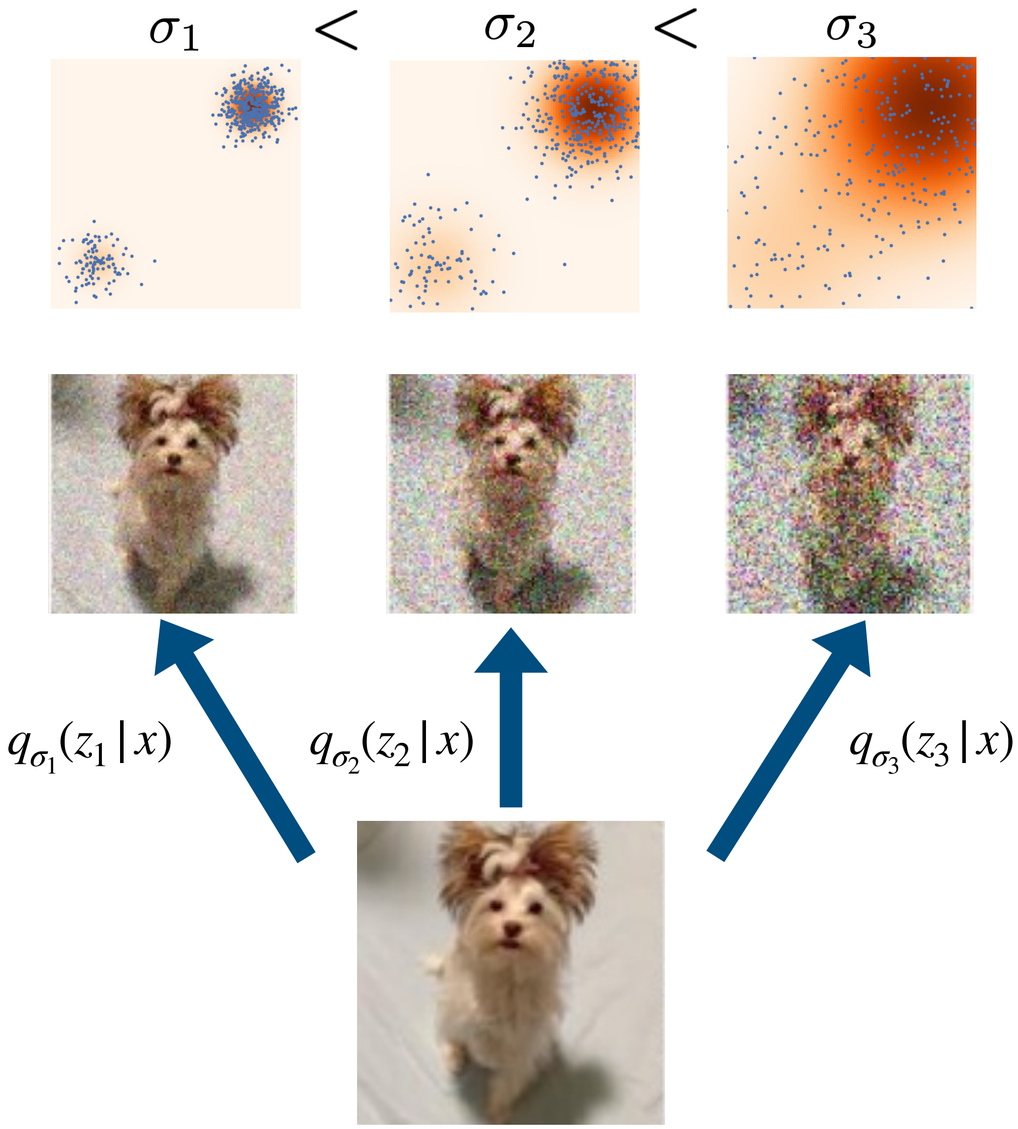




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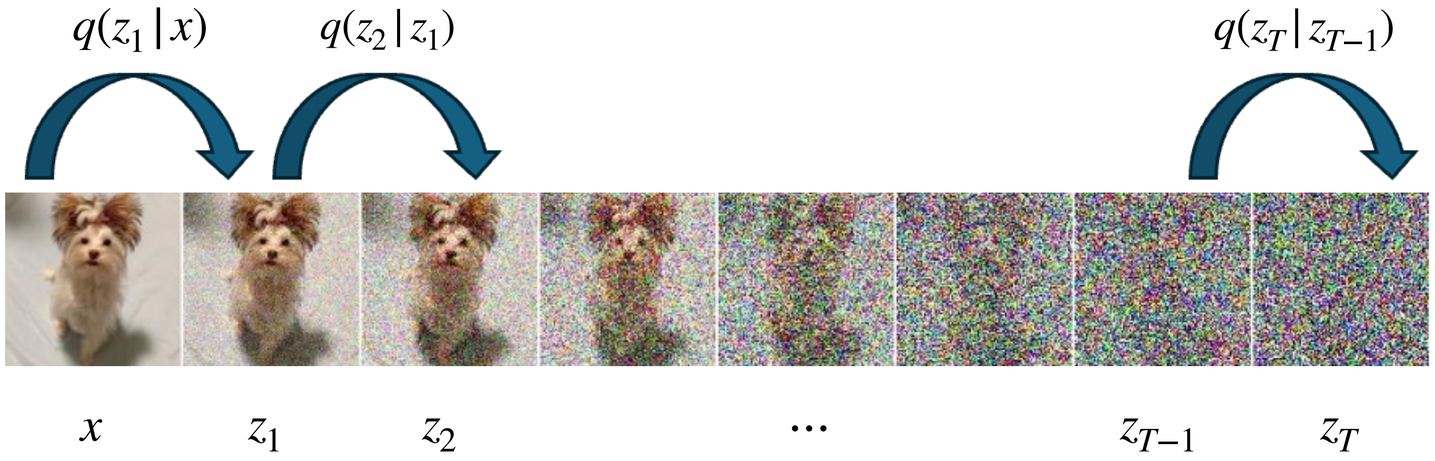
Image Source: Blog on Score-based Generative Models — Yang Song

Noising Process



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Iterative Noising Process



- Slowly destroy structure in an iterative process by adding Gaussian noise:

$$q(z_t \mid z_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t} z_{t-1}, \beta_t \mathbf{I}), \qquad \{\beta_t \in (0,1)\}_{t=1}^{T}$$

- Defines a Markov chain with joint distribution:

$$q(z_{1:T} \mid x) = \prod_{t=1}^{T} q(z_t \mid z_{t-1})$$

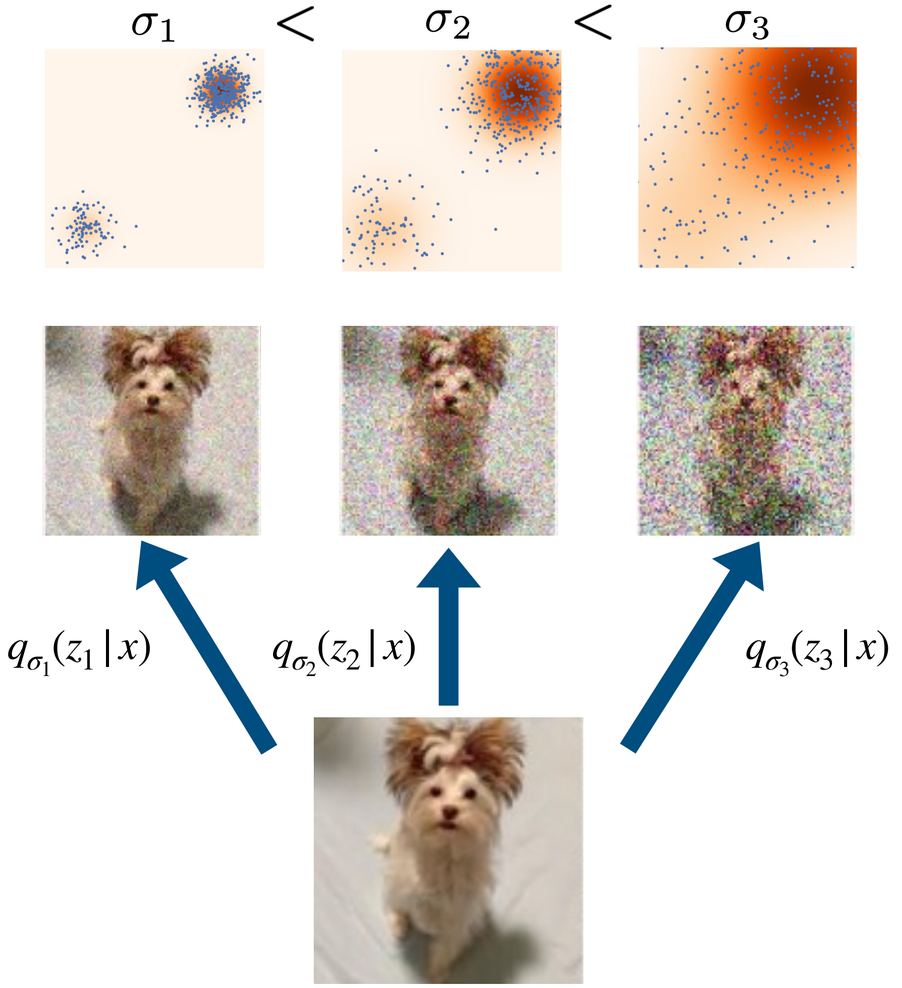
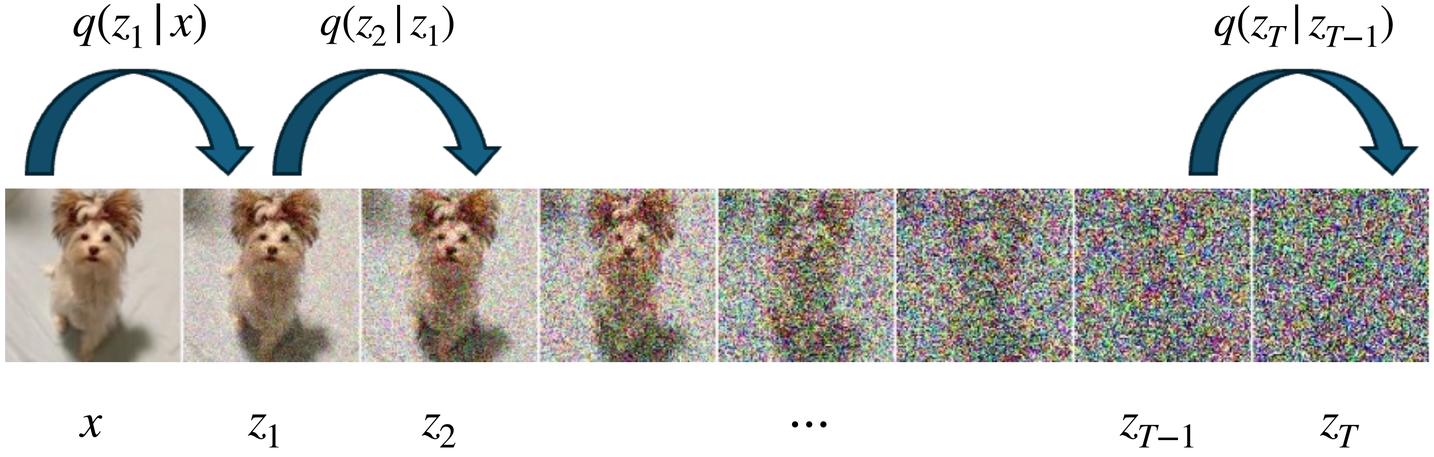


Image Source: Blog on Score-based Generative Models — Yang Song

Multistep Transition



Diffusion Models

$$z_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$$

Denoising Score Matching

$$z_t = x_0 + \sigma_t \epsilon$$

- Multistep transitions are also Gaussian can be computed in close form:

$$q(z_t \mid x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)\mathbf{I}), \qquad \bar{\alpha}_t = \prod_{s=1}^{t} (1 - \beta_s)$$

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Diffusion Perspective

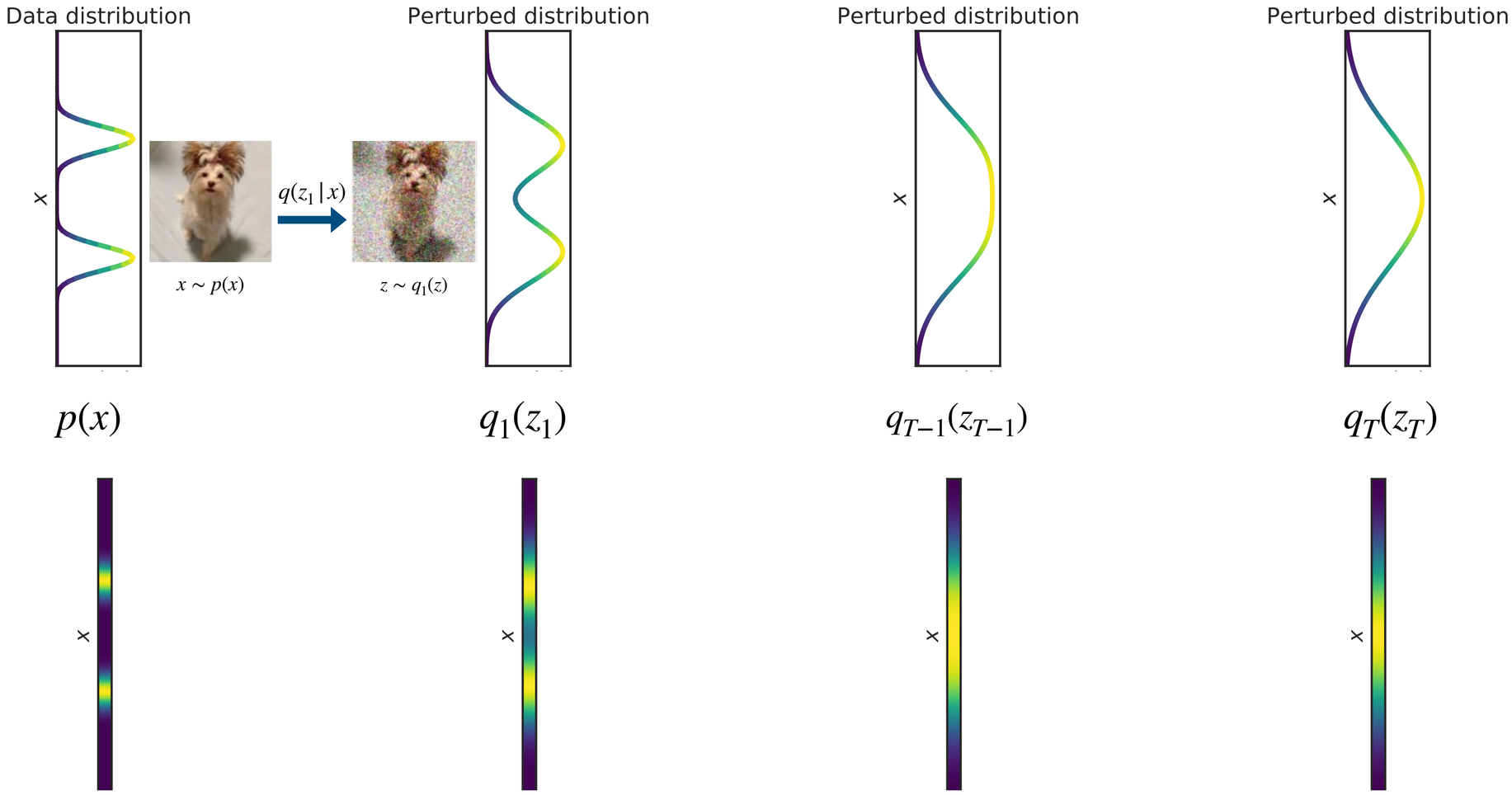
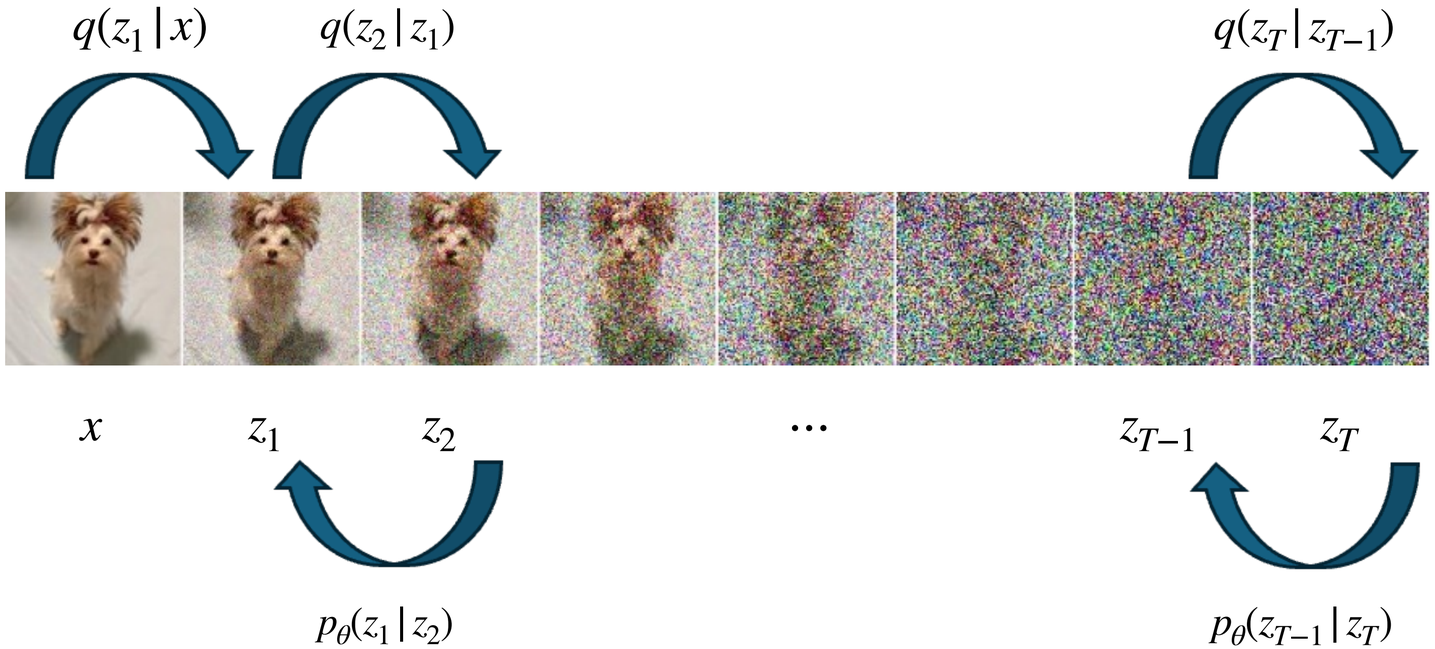


Image Source: From slides of CS236 Deep Generative Models lecture

Iterative Denoising



Sampling:

1. Sample $z_T \sim \mathcal{N}(0, \mathbf{I})$
2. Iteratively sample from $p_\theta(z_{t-1} \mid z_t)$

- During training we want to learn a reverse process that restores structure:

$$p_\theta(z_{t-1} \mid z_t) = \mathcal{N}(\mu_\theta(z_t, t), \beta_t \mathbf{I})$$

- Joint distribution:

$$p_\theta(x, z_{1:T}) = p(z_T) \prod_{t=2}^{T} p_\theta(z_{t-1} \mid z_t)\ p_\theta(x \mid z_1)$$

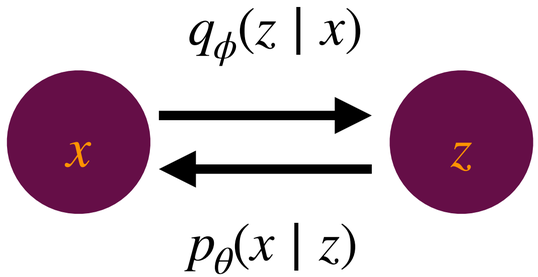
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Diffusion Model as Hierarchical VAE

Variational Autoencoder (VAE)

- One layer of latent z.

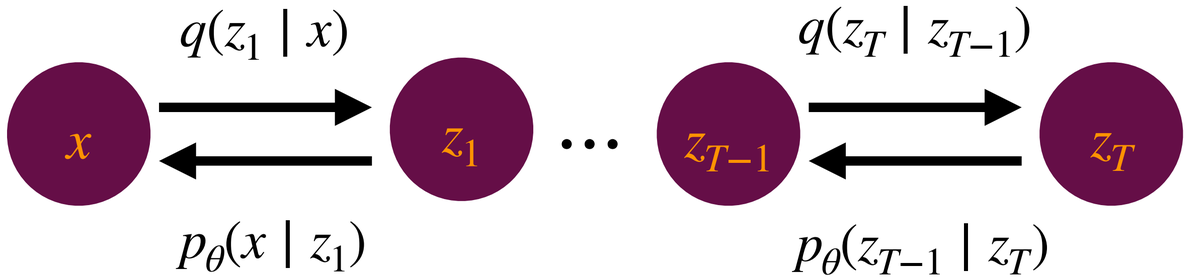
Evidence Lower Bound (ELBO): $\mathbb{E}_{q_\phi(z|x)}\left[\log \frac{p_\theta(x,z)}{q_\phi(z\,|\,x)}\right]$



Hierarchical VAE (HVAE)

- Multiple layers of latents.
- In Markovian HVAE, each latent is generated from only the previous latent.

Evidence Lower Bound (ELBO): $\mathbb{E}_{q(z_{1:T}|x)}\left[\log \frac{p_\theta(x,z_{1:T})}{q(z_{1:T}\,|\,x)}\right]$



Diffusion model is an instance of HVAE where

$$q(z_t\,|\,z_{t-1}) = \mathcal{N}(z_t \mid \sqrt{1-\beta(t)}z_{t-1},\, \beta(t)\mathbf{I})$$

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Computing the ELBO

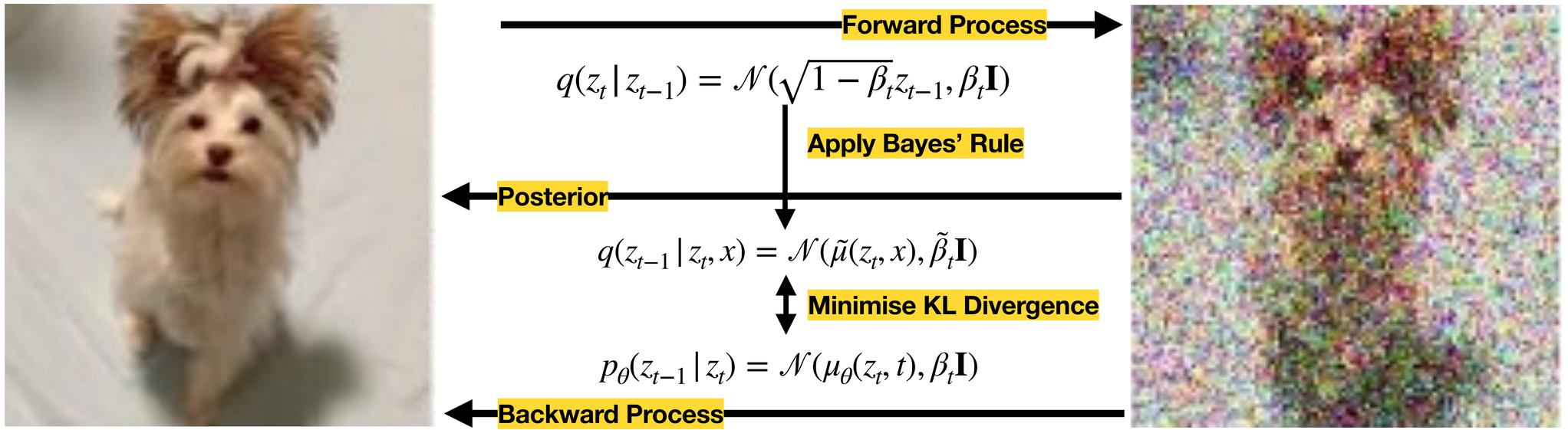
$$\mathbb{E}_{q(z_{1:T}|x)}\left[\log \frac{p_\theta(x, z_{1:T})}{q(z_{1:T}\,|\,x)}\right]$$

$$= \mathbb{E}_{q(z_{1:T}|x)}[\,\mathrm{KL}(q(z_T \mid x)\|p(z_T)) + \sum_{t=2}^{T} \mathrm{KL}(q(z_{t-1} \mid z_t)\|p_\theta(z_{t-1} \mid z_t)) - \log\ p_\theta(x \mid z_1)]$$

- First term is **prior matching term** does not depend on θ.
- Second term is the **consistency term.**
  - KL divergence between two Gaussians with the same variance.
  - Closed form solution.
- Third term is the **reconstruction term** similar as in VAEs.

Computing the ELBO

$$\mathrm{KL}(q(z_{t-1} \mid z_t) \| p_\theta(z_{t-1} \mid z_t))$$



z_{t-1} z_t

$$\tilde{\mu}(z_t, x) = \frac{\sqrt{1-\beta_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} z_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} x \quad \text{and} \quad \tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$$

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Computing the ELBO

$$\mathrm{KL}(q(z_{t-1} \mid z_t) \| p_\theta(z_{t-1} \mid z_t))$$

- KL Divergence between two Gaussians can be computed as:

$$\mathrm{KL}(q(z_{t-1} \mid z_t) \| p_\theta(z_{t-1} \mid z_t)) = \frac{1}{2\sigma_t^2} \| \tilde{\mu}_t(z_t, x) - \mu_\theta(z_t, t) \|^2$$

- We can directly use this to train a model that predicts the mean.
- However, this was found to be not working well in practice.

Parameterising the noise

$$\mathrm{KL}(q(z_{t-1} \mid z_t) \| p_\theta(z_{t-1} \mid z_t)) = \frac{1}{2\sigma_t^2} \| \tilde{\mu}_t(z_t, x) - \mu_\theta(z_t, t) \|^2$$

- Recall the mean we are trying to learn is:

$$\tilde{\mu}(z_t, x) = \frac{\sqrt{1-\beta_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} z_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} x$$

- Here, $z_t = \sqrt{\bar{\alpha}_t} x + \sqrt{1-\bar{\alpha}_t}\epsilon$. Using this we can rewrite the mean as

$$\tilde{\mu}(z_t, x) = \frac{1}{\sqrt{1-\beta_t}} \left(z_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon \right)$$

$$\mu_\theta(z_t, x) = \frac{1}{\sqrt{1-\beta_t}} \left(z_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t) \right)$$

- Ho et al. proposed to learn the noise prediction network. Thus

$$\mathbb{E}_{x \sim q(x), \epsilon \sim \mathcal{N}(0, \mathbf{I})} \left[\lambda_t \| \epsilon - \epsilon_\theta(\sqrt{\bar{\alpha}_t} x_0 + \sqrt{1-\bar{\alpha}_t}, t) \|^2 \right]$$

Training and Inference

- The final loss is obtained by taking an expectation over the steps

$$\mathbb{E}_{x\sim q(x),t\sim[1,T],\epsilon\sim\mathcal{N}(0,\mathbf{I})}\left[\|\epsilon-\epsilon_\theta(\sqrt{\bar{\alpha}_t}x_0+\sqrt{1-\bar{\alpha}_t},t)\|^2\right]$$

Algorithm 1 Training

1: **repeat**
2: $\mathbf{x}_0 \sim q(\mathbf{x}_0)$
3: $t \sim \text{Uniform}(\{1,\dots,T\})$
4: $\boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0},\mathbf{I})$
5: Take gradient descent step on
$\nabla_\theta \left\|\boldsymbol{\epsilon}-\boldsymbol{\epsilon}_\theta(\sqrt{\bar{\alpha}_t}\mathbf{x}_0+\sqrt{1-\bar{\alpha}_t}\boldsymbol{\epsilon},t)\right\|^2$
6: **until** converged

Algorithm 2 Sampling

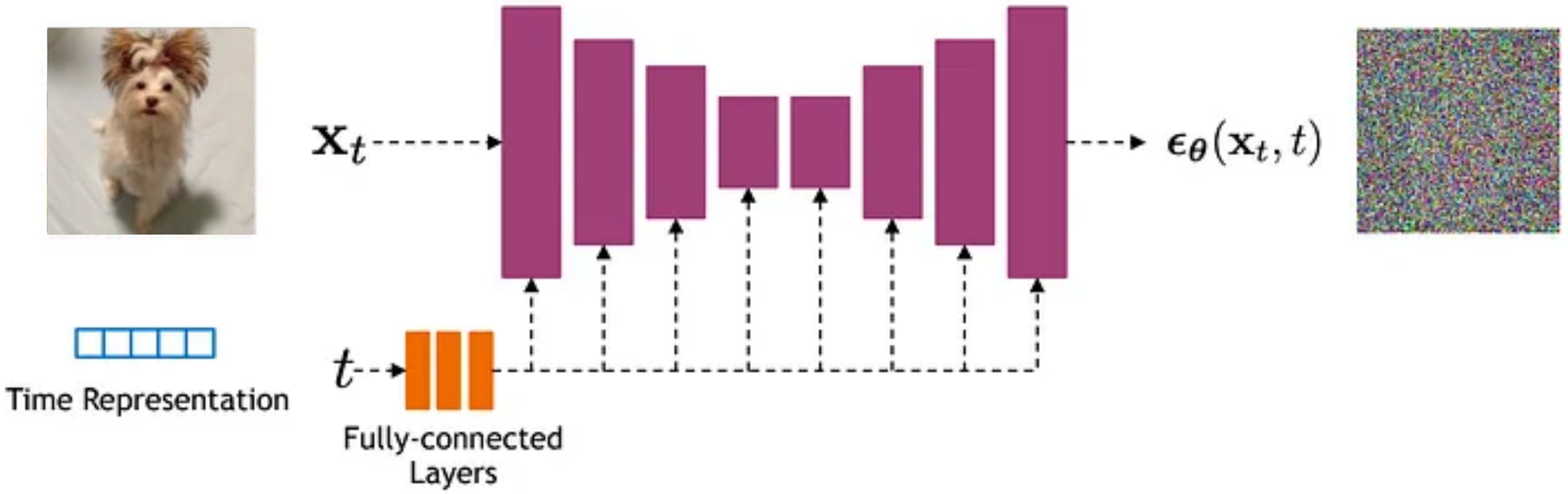
1: $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0},\mathbf{I})$
2: **for** $t=T,\dots,1$ **do**
3: $\mathbf{z}\sim\mathcal{N}(\mathbf{0},\mathbf{I})$ if $t>1$, else $\mathbf{z}=\mathbf{0}$
4: $\mathbf{x}_{t-1}=\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t-\frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\boldsymbol{\epsilon}_\theta(\mathbf{x}_t,t)\right)+\sigma_t\mathbf{z}$
5: **end for**
6: **return** $\mathbf{x}_0$

Denoising Score Matching

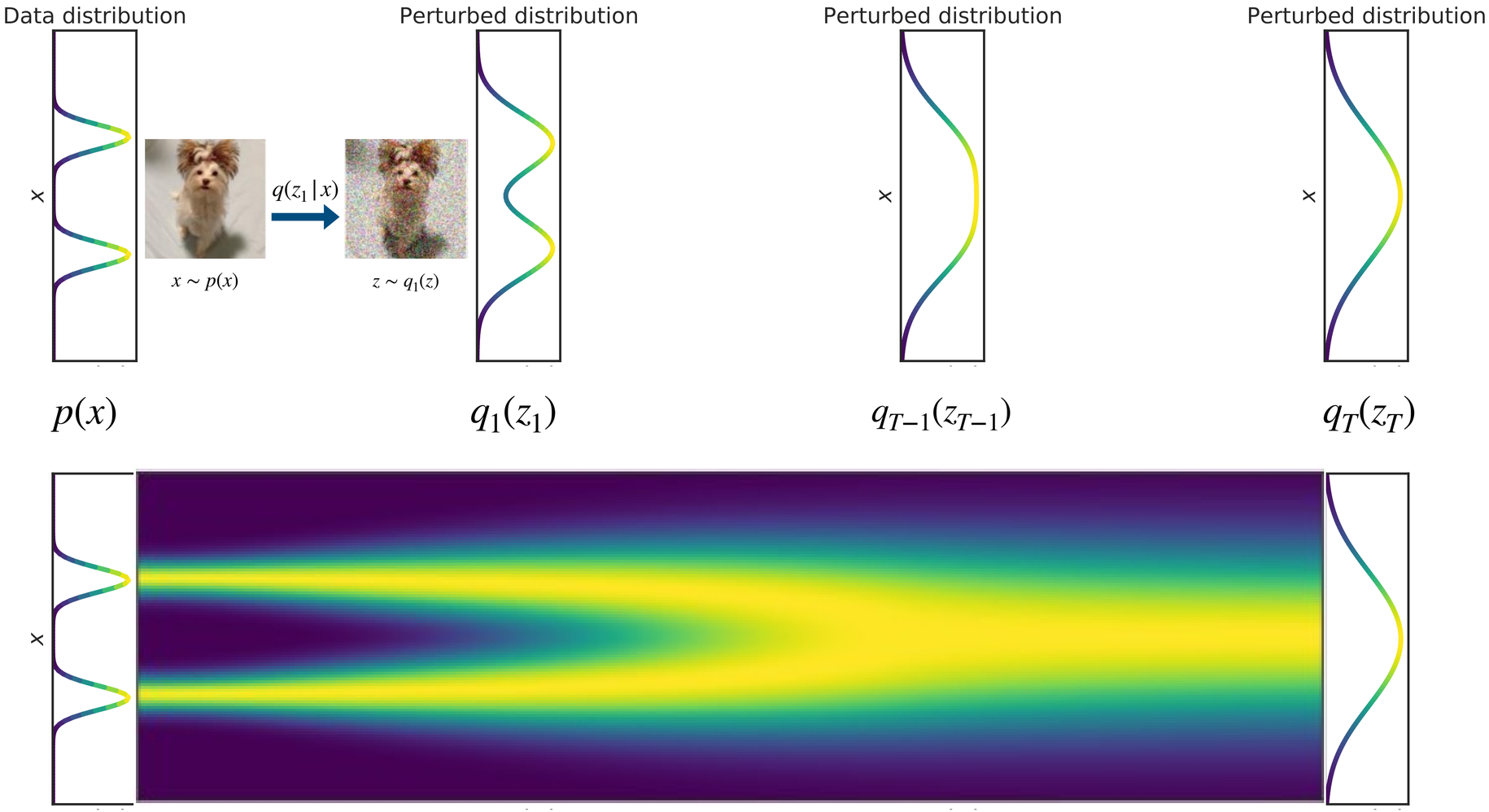
Annealed Langevin Dynamics

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Diffusion Network



Infinite Noise Scales

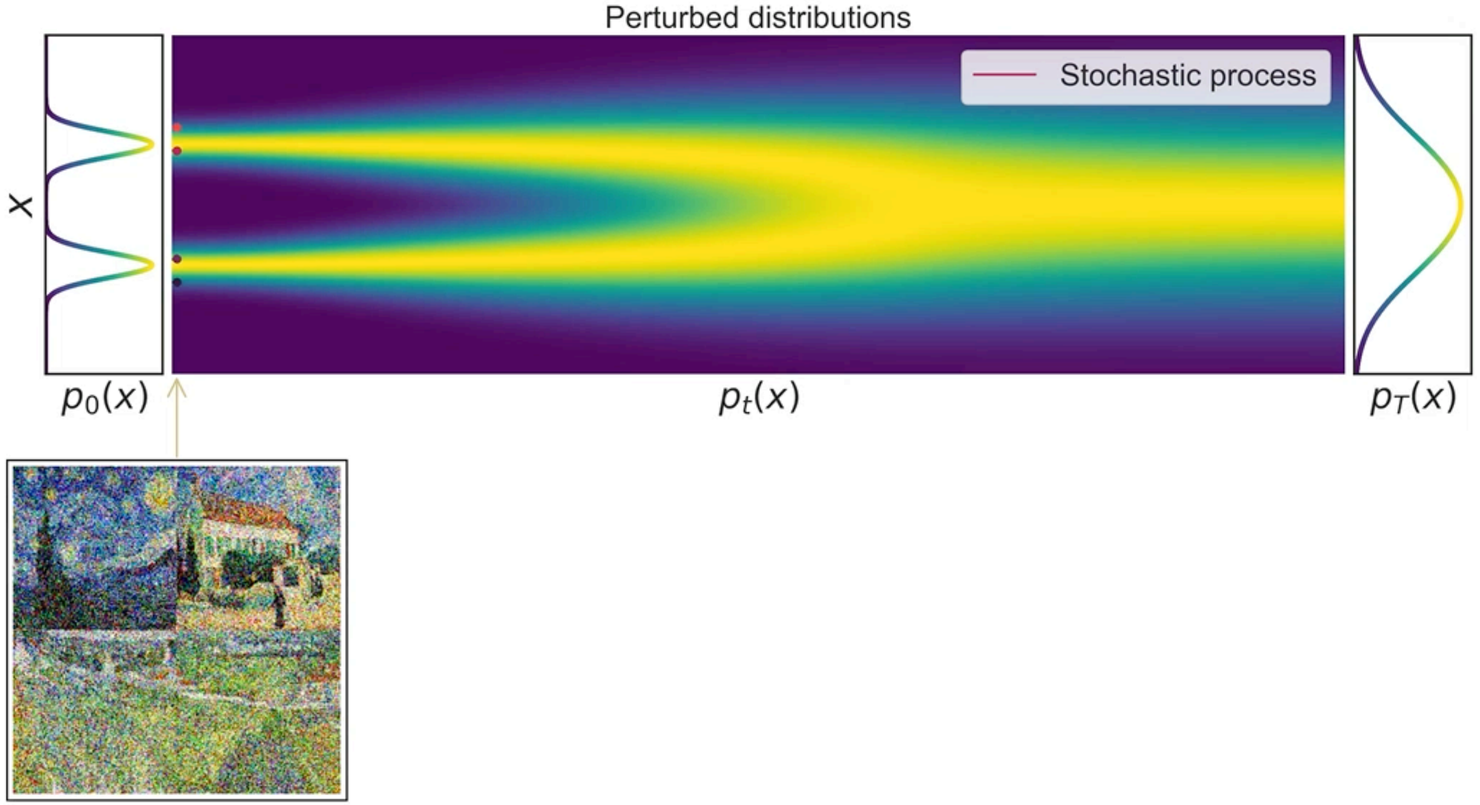


Continuous vs Discrete Diffusion

	Discrete Diffusion	Continuous Diffusion
Timesteps	$t \in \{1, \cdots, T\}$	$t \in [0,1]$
Noise levels	$\sigma_t \in \{\sigma_1, \cdots, \sigma_T\}$	$\sigma : [0,1] \to \mathbb{R}_{\geq 0}$
Diffused samples	$z_t \in \{z_1, \cdots, z_T\}$	$x : [0,1] \to \mathbb{R}^d$
	$z_t = x + \sigma_t \epsilon$	$x(t) = x(0) + \sigma(t)\epsilon$

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Perturbing with Stochastic Process



Stochastic Differential Equation (SDE)

$$dx = f(x,t)dt + g(t)dw_t$$

Drift coefficient **Diffusion coefficient**

Diffusion Models

$$x(t) = \sqrt{\bar{\alpha}_t}x(0) + \sqrt{1-\bar{\alpha}_t}\epsilon$$

$$dx = -\frac{1}{2}\beta(t)xdt + \sqrt{\beta(t)}dw_t$$

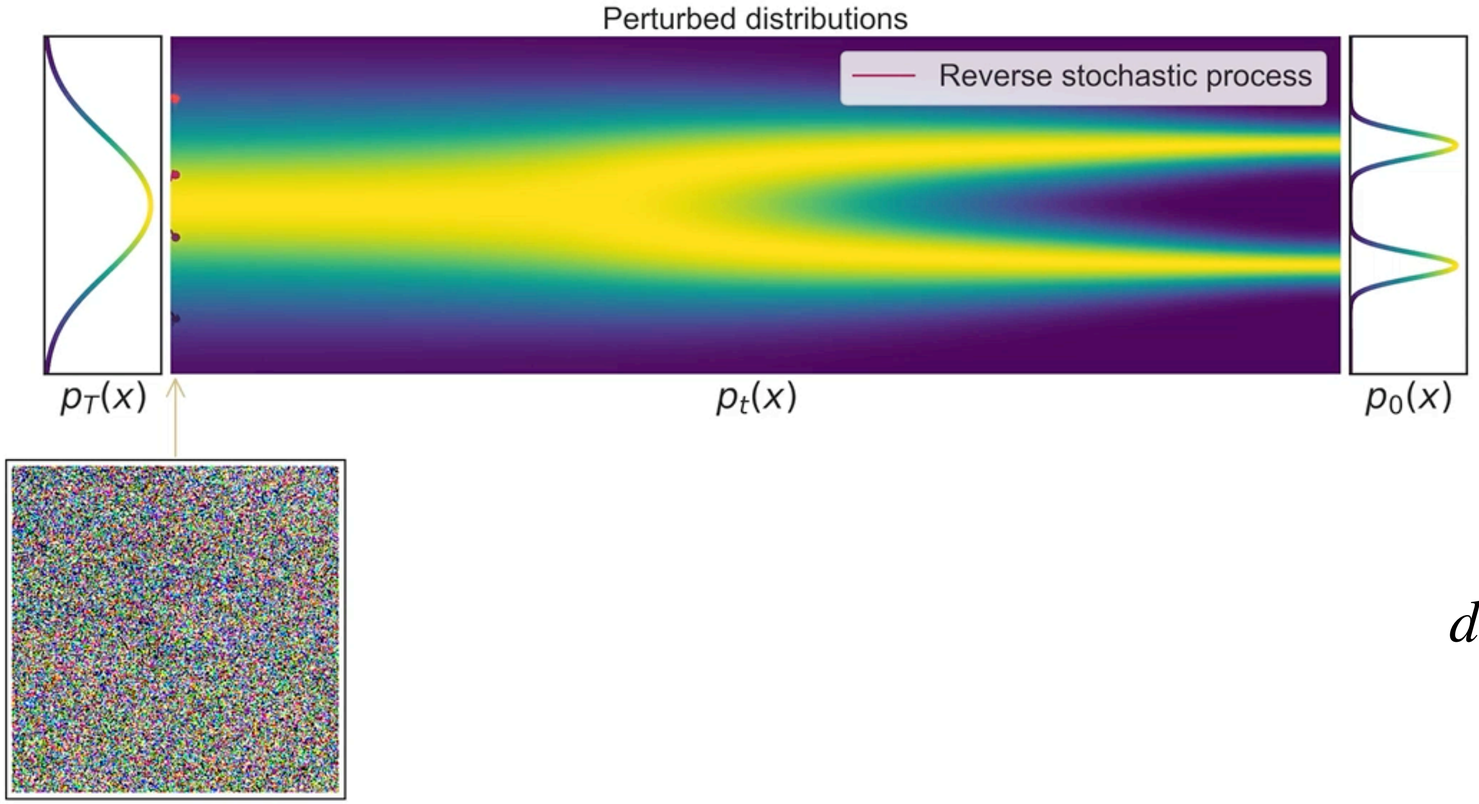
Denoising Score Matching

$$x(t) = x(0) + \sigma(t)\epsilon$$

$$dx = \sqrt{\frac{d[\sigma(t)]}{dt}}dw_t$$

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Generation via Reverse Stochastic Process



Stochastic Differential Equation (SDE)

$$dx = f(x,t)dt + g(t)dw_t$$

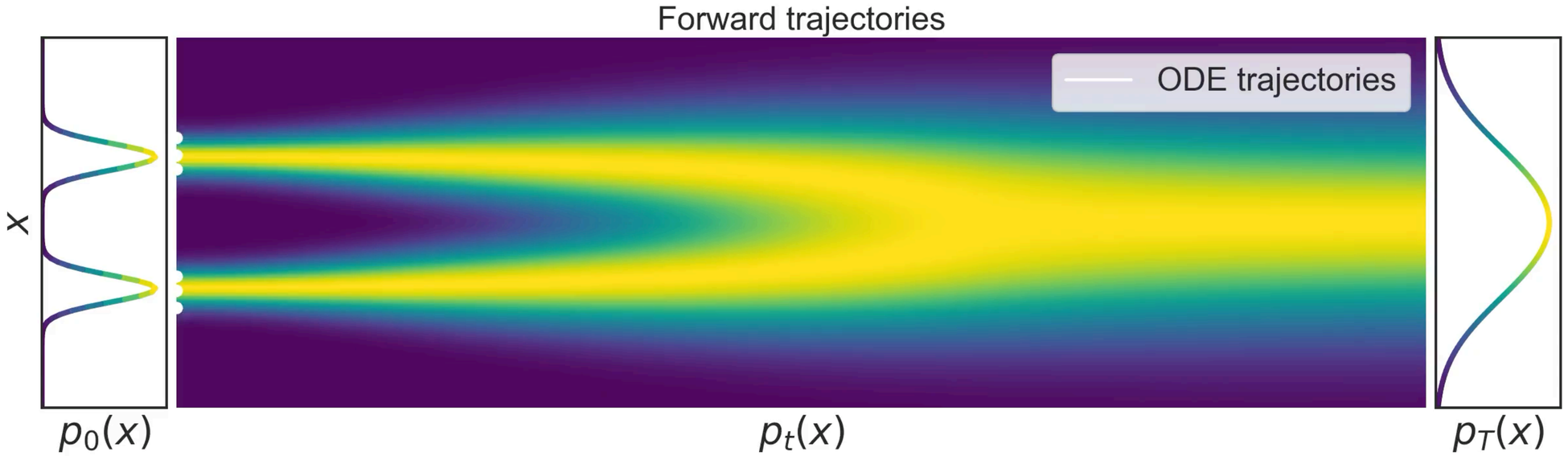
Anderson, 1982

Reverse SDE

$$dx = [f(x,t)dt - g(t)^2 \nabla_x p_t(x)]dt + g(t)dw_t$$

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From SDE to ODE



Forward SDE

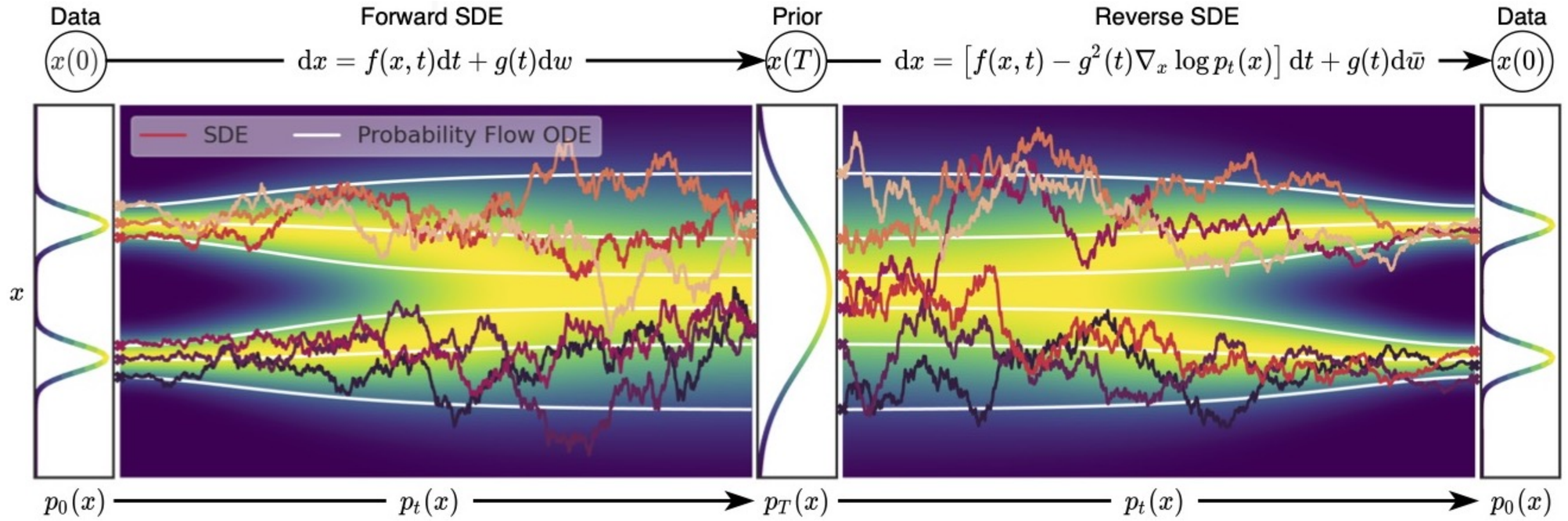
$$dx = f(x,t)dt + g(t)dw_t$$

Forward Ordinary Differential Equation (ODE)

$$dx = [f(x,t) - \frac{1}{2}g(t)^2 \nabla_x \log\ p_t(x)]dt$$

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Probability Flow ODE

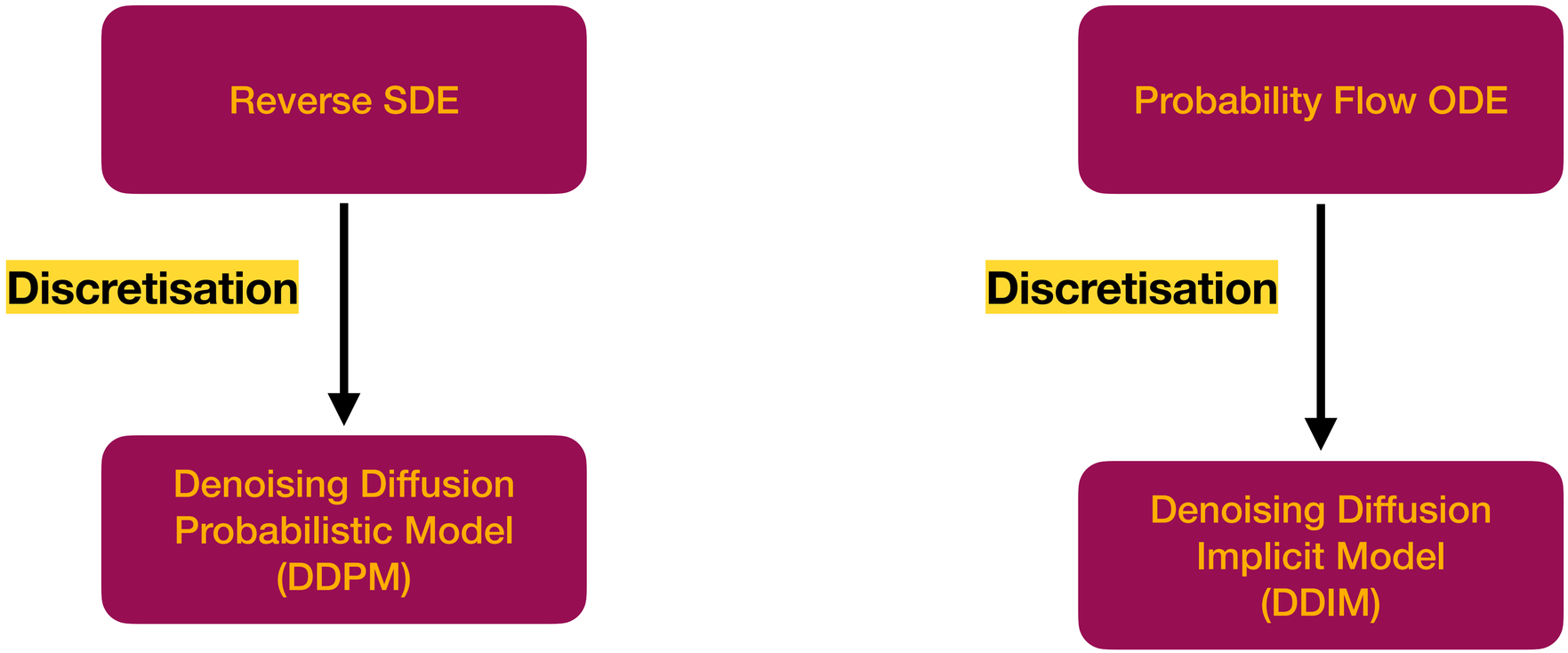


Think of this as a (continuous time, infinite depth) normalising flow.

- Unique ODE solution → results in a deterministic, reversible process.
- Allows computation of the likelihood using the change of variable formula.

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High-level View



Summary

1. Iterative Noising and Denoising
2. Denoising Diffusion Models
3. Continuous Diffusion Models



“cat scientist in a classroom with a blackboard that says “Thank you!”, digital art.”

Flux.1 Merged

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